2022-Summer Paper 1 – Basic Sciences in Urology

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Q1. Describe anatomy of kidney in relation to PCNL and describe indication, contraindication, procedure and complications of mini-PCNL.

Type: LAQ (25 marks) (also answered in 2017-Winter Paper 1, Q1 (pelvicalyceal anatomy) and 2017-Winter Paper 4, Q3d (Miniperc))

Introduction

- Safe PCNL (percutaneous nephrolithotomy) access depends on the kidney's position, neighbours, vessels and calyces.

Surgical anatomy relevant to PCNL

1. Position and orientation

- Lies T12–L3 (right 1–2 cm lower); descends about 3 cm on inspiration.
- It lies on the cone-shaped psoas, so the upper pole is medial and posterior and the lower pole lateral and anterior.
- Rotated 30–50° behind the coronal plane, so **posterior calyces point posterolaterally**, towards the prone surgeon.
- Tract: skin, flank muscles, thoracolumbar fascia, paranephric fat, Gerota fascia, perirenal fat, cortex, then **pyramid and papilla** into the calyx.

2. Relations (the hazards of access)

Structure	Relation	Practical point
Pleura, lung	Pleura reaches the 12th rib; lung the 10th intercostal space	Prefer subcostal or 11th-space access; risk rises above the 12th rib, highest above the 11th; puncture in expiration
Liver, spleen	Over the upper poles; descend in inspiration	Risk with supracostal access, inspiration, organomegaly
Colon	Retrorenal colon, mainly at the lower pole, commoner prone and on the left	Review CT; ultrasound shows bowel

3. Vascular anatomy

- Hilum, anterior to posterior: vein, artery, pelvis, **posterior segmental artery** (only branch behind the pelvis).

- Five segmental **end arteries**; interlobar arteries run in the columns of Bertin beside the infundibula.
- **Brödel line:** avascular plane between anterior and posterior territories, just posterior to the lateral border. Posterior calyces along it are the target.

Puncture site (Sampaio)	Vascular injury
Upper-pole infundibulum	67% (arterial 26%)
Mid-pole infundibulum	Arterial 23%
Lower-pole infundibulum	Infundibular artery present in 38%
Calyceal fornix	Venous <8%, no arterial injury

- So the needle passes **through the papilla along the calyx axis**, never into an infundibulum or the pelvis.

4. Pelvicalyceal system

- 7–9 calyces: upper pole 3, interpolar 3–4, lower pole 2–3; anterior and posterior rows overlap on a pyelogram.
- **Sampaio types:** A-I 45%, A-II (crossed calyces) 17.2%, B-I 21.4%, B-II 16.4%; perpendicular minor calyx 11.4%.
- Ideal target: a **posterior calyx with a short, wide infundibulum**, identified by injected air (rises posteriorly when prone), C-arm rotation or ultrasound. Upper-pole access reaches the UPJ (ureteropelvic junction) and staghorn branches best; lower pole is chest-safe.
- **Also know:** horseshoe (medial access), pelvic and transplant kidneys.

Mini-PCNL: definition

Term	Tract size
Standard	24–34 Fr
Mini	15–22 Fr
Ultra-mini	11–14 Fr
Micro-perc	<10 Fr

- First mini-perc: Jackman, 1998 (13 Fr).

Indications

- Renal stones 1–2 cm; lower-pole stones >1 cm; failed SWL (shock wave lithotripsy) or RIRS (retrograde intrarenal surgery).
- Calyceal diverticulum (lower pole), children, poor retrograde access (urinary diversion, ureteric stricture), selected stones >2 cm or extra tracts for staghorns.

Contraindications

- Untreated UTI (urinary tract infection), uncorrected coagulopathy, pregnancy (EAU, European Association of Urology).
- Tumour in the access tract or suspected malignant renal tumour; no safe window; unfit for anaesthesia.

Procedure

1. Urine culture, blood count, coagulation, creatinine; **non-contrast CT (computed tomography)**; antibiotic prophylaxis.
2. Lithotomy: ureteric catheter and retrograde pyelogram.
3. Prone, or supine (Galdakao-modified) for single-position retrograde access.

4. 18 G puncture of a posterior calyx under fluoroscopy (bull's-eye or triangulation) or ultrasound; urine return; guidewire and safety wire.
5. One-step, serial or balloon dilatation to 15–22 Fr; place the sheath.
6. Mini-nephroscope; holmium/thulium laser or pneumatic lithotripsy. Fragments wash out by the irrigation **"vacuum-cleaner" effect** or a **suction sheath**.
7. Flexible nephroscopy to clear calyces.
8. Exit: tubeless, small nephrostomy (≤10 Fr: less pain) or double-J stent.

Advantages and limitations

- Meta-analysis: **non-inferior stone-free rate**, smaller haemoglobin drop, **fewer transfusions**, shorter stay.
- Longer operation, poorer vision. A tight sheath raises **intrarenal pressure**, causing pyelovenous backflow and **fever or sepsis**. Advantages disappear against 24 Fr tracts.

Complications (grade by Clavien–Dindo)

Type	Points
Bleeding	Acute tract bleeding: tamponade sheath, balloon or Kaye catheter. Delayed: pseudoaneurysm or AV (arteriovenous) fistula, treated by angioembolisation
Infection	Fever, SIRS (systemic inflammatory response syndrome), urosepsis
Thoracic	Pneumothorax, hydrothorax: chest drain
Visceral	Colon (stent plus nephrostomy withdrawn into the colon), spleen, liver, duodenum
Collecting system	Perforation, extravasation, later infundibular stenosis; residual stones
Others	Irrigation absorption, venous gas embolism, DVT (deep vein thrombosis), neuropraxia, renal function loss, death

Diagram to draw: transverse section of kidney showing posterior calyx, Brödel line and the needle through the fornix.

> **Key points to remember:**
>
> - Fornix puncture: no arterial injury; upper-pole infundibulum 67% vascular injury.
> - Pleura to 12th rib; retrorenal colon at the lower pole, commoner prone.
> - Mini 15–22 Fr (Jackman 1998): fewer transfusions, longer operation.
> - High intrarenal pressure in mini-PCNL causes fever and sepsis.

Q2. Urological evaluation and management of pediatric kidney transplant patient.

Type: LAQ (25 marks)

Introduction

- Children needing KRT (kidney replacement therapy) often have CAKUT (congenital anomalies of the kidney and urinary tract). The urologist must ensure that the bladder that destroyed the native kidneys does not destroy the graft.
- **Five goals:** unobstructed drainage; a capacious, low-pressure reservoir with social continence; volitional emptying; no infection; the fewest operations.

Causes of ESKD (end-stage kidney disease) in children

- Commonest age at transplant 13–17 years.
- **Younger:** dysplasia, hypoplasia, aplasia and obstructive uropathy, mainly PUV (posterior urethral valves). **Older:** FSGS (focal segmental glomerulosclerosis) and neuropathic bladder.

Pretransplant evaluation

- **Every child:** urological history, renal ultrasound, uroflow with PVR (post-void residual) if toilet-trained.
- **VCUG (voiding cystourethrogram)** only with neuropathic bladder, PUV, VUR (vesicoureteral reflux), previous lower-tract surgery, recurrent febrile UTI (urinary tract infection) or unexplained hydronephrosis.
- **Urodynamics** for neuropathic bladder, PUV or large ureterocele, refractory voiding dysfunction, high-grade hydronephrosis or recurrent UTI. They measure capacity, compliance, detrusor activity, emptying and sphincter function, and are repeated.

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History positive → ultrasound + VCUG → urodynamics
  Poor capacity/compliance → anticholinergic, CIC, cycling → still poor → augment
  Poor emptying → CIC ± catheterisable stoma
  Adequate → clear for transplant
History negative → ultrasound + PVR → normal → clear
```

- An anuric child with no urological history needs no further tests.

Pretransplant management

1. Bladder optimisation (drugs → CIC → cycling → augmentation)

- Commonest problem: low capacity, detrusor overactivity, poor compliance. High pressure harms the kidney even without reflux.
- **Target:** age-expected capacity at detrusor pressure **<40 cm H_2O**.
- Antimuscarinics (oxybutynin titrated; solifenacin), mirabegron, intradetrusor botulinum toxin; α-blockers aid emptying. Better storage may worsen emptying, so add **CIC (clean intermittent catheterisation)**. Prove volitional emptying by repeated low PVRs.
- A family's adherence to CIC predicts adherence to immunosuppression.

2. Valve bladder versus neuropathic bladder

- PUV bladder moves from overactivity to **myogenic failure**: postpone augmentation. Neuropathic bladder may deteriorate under high post-transplant urine output: reconstruct earlier.

3. Defunctionalised bladder (anuria or diversion)

- **Bladder cycling:** catheter instillation of increasing saline volumes, with drugs, until age-expected capacity; confirm by urodynamics. **Never use the graft to test the bladder.**

4. Augmentation

- Indication: cannot reach **>75% expected capacity at <40 cm H_2O** with CIC every 3–4 hours despite maximal therapy. It does not clearly raise transplant complications.

Segment	Point
Dilated ureter (ureterocystoplasty)	Explore first: urothelium, no mucus or acidosis
Ileum, sigmoid	Most used; mucus, stones, infection, malignancy; lifelong follow-up
Stomach	Haematuria–dysuria syndrome in anuric patients; rarely used

- Continent stoma (Mitrofanoff) if the urethra is sensate or difficult; site it medial to the planned incision. Allow **6–12 weeks** healing before transplant.

5. Infection, nephrectomy and dialysis

- Treat stasis (double voiding, CIC); distinguish colonisation from UTI; consider circumcision in boys with valves.
- **Native nephrectomy** (default is to keep): uncontrolled hypertension, nephrotic syndrome, massive polyuria, recurrent pyelonephritis, large stones, malignancy risk, very large kidneys, massive VUR. Retroperitoneal route keeps PD (peritoneal dialysis).
- Keep a normal native ureter for later ureteroureterostomy. Intraperitoneal surgery on PD needs temporary haemodialysis.

At transplantation

- Retroperitoneal Gibson incision; children **<15 kg** may need aorta–caval anastomosis, sometimes transperitoneal.
- **Ureter:** extravesical non-refluxing ureteroneocystostomy (3–4 cm detrusorotomy) preferred; Barry tunnel; ureteroureterostomy to a normal native ureter.
- **Stent:** selective; double-J removed at 4–6 weeks, or early removal at 5–7 days.

Post-transplant complications

Complication	Rate	Management
Ureteric obstruction	6%	Usually distal ischaemic stenosis; >half within 100 days. Stent or nephrostomy; balloon dilatation; open repair (native ureter, bladder flap)
Urine leak	3%	Early; drain creatinine; catheter, nephrostomy, stent, then reimplantation
Graft VUR	2% (significant)	Febrile UTI → reimplant; asymptomatic → observe; injection works in one-third
Stones	2%	Treat if >1 cm and growing, obstructing or infected

- **Rising creatinine with hydronephrosis:** catheter drainage 1–2 weeks (bladder cause?) → biopsy if rejection is suspected → stent or nephrostomy. Obstruction and rejection can coexist.
- Also: lymphocele, **PTLD (post-transplant lymphoproliferative disorder)**, graft thrombosis (a leading cause of early graft loss), late infection reflecting bladder dysfunction.

Follow-up

- Lifelong ultrasound, PVR and bladder surveillance; recurrent UTIs lower graft GFR (glomerular filtration rate).

Diagram to draw: extravesical ureteroneocystostomy with detrusor flaps closed over the ureter.

Key points to remember:

- Bladder target: age-expected capacity at <40 cm H_2O; augment if <75% despite CIC.
- PUV bladder heads to myogenic failure; neuropathic bladder needs earlier reconstruction.
- Never test a bladder with a graft; cycle the defunctionalised bladder.
- Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.

Q3a. Instestinal segments in urology

Type: SAQ (10 marks)

Assumed meaning: intestinal segments used in urinary diversion and bladder reconstruction.

Introduction

- Bowel is used for conduits, continent pouches, neobladders, augmentation and ureteric replacement. Choice depends on renal function, irradiation, previous surgery, mesenteric length and the reconstruction needed.

Segments compared

Segment	Electrolyte disturbance	Advantage	Drawback
Stomach	Hypochloraemic, hypokalaemic metabolic alkalosis	Secretes acid, less mucus; for renal failure, short gut	Haematuria–dysuria syndrome, ulcers
Jejunum	Hyponatraemic, hypochloraemic, **hyperkalaemic** acidosis	Outside radiation field	Severe salt loss; avoid
Ileum	**Hyperchloraemic metabolic acidosis**	Mobile, constant supply; default conduit	B12 and bile-salt loss; obstruction ~10%
Colon	Hyperchloraemic metabolic acidosis	Easy antireflux tunnel; transverse colon spared by pelvic radiation	Diarrhoea if ileocaecal valve used

- **Mechanism (ileum, colon):** NH_4^+ replaces Na^+ on the Na^+/H^+ exchanger, coupled with Cl^-/HCO_3^- exchange, so ammonium chloride is absorbed. Treat with sodium bicarbonate or citrate (potassium citrate if hypokalaemic); chlorpromazine or nicotinic acid limit chloride transport.

Other complications

- **Hypokalaemia** (worse with colon); correct potassium before acidosis.
- **Altered sensorium:** hyperammonaemia (drain diversion, lactulose), hypomagnesaemia, drug reabsorption (phenytoin, methotrexate).
- **Osteomalacia:** acidosis, vitamin D resistance, calcium loss; growth retardation in children.
- **Infection, stones** (ileal conduit 10–12%, continent caecal reservoir ~20%), short-gut effects.
- **Cancer:** ureterosigmoidostomy 6–29% after 10–20 years; endoscopy from 5 years.

Renal function needed for continent diversion

- Creatinine <2 mg/dL; GFR (glomerular filtration rate) >35 mL/min; urine pH ≤5.8 after ammonium chloride load; osmolality ≥600 mOsm/kg.

Technical principles

- Clear ≤8 cm of mesentery from a segment end; avoid watershed areas (Griffiths, Sudeck points).
- **Also know:** ERAS (enhanced recovery after surgery) with no routine mechanical bowel preparation; Wallace (stricture ~3%) and Bricker anastomoses; nipple stoma.

Diagram to draw: ileal loop with mesenteric arcade and Wallace anastomosis.

Key points to remember:

- Ileum and colon: hyperchloraemic acidosis (NH_4Cl absorption).
- Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
- Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.

Q3b. Burch colposuspension

Type: SAQ (10 marks) (related: 2017-Winter Paper 3, Q2)

Definition

- Retropubic suspension (**Burch, 1961**) of the paraurethral and para-bladder-neck vaginal wall to **Cooper's (iliopectineal) ligament**; Tanagho (1976) modified it.

Principle

- SUI (stress urinary incontinence) from **urethral hypermobility**: the urethra drops out of the retropubic position. Elevation restores **pressure transmission** and a backstop; scarring holds it.
- It treats hypermobility, not ISD (intrinsic sphincter deficiency).

Indications

- SUI or stress-predominant MUI (mixed urinary incontinence) with hypermobility, especially with **concomitant abdominal surgery** (sacrocolpopexy), **mesh refusal**, or poor vaginal access.
- **Contraindications:** immobile urethra or vagina, ISD alone, pure urgency incontinence.

Work-up

- Cough stress test, PVR (post-void residual), urinalysis; urodynamics only if SUI is not demonstrated.

Steps (open)

1. Low lithotomy, Foley; Pfannenstiel incision; enter the **space of Retzius**.
2. Vaginal finger elevates the vagina; sweep bladder medially. **Avoid midline dissection** over urethra.
3. Two to four 0 or 1-0 delayed-absorbable or non-absorbable sutures each side through vaginal wall (not lumen): distal at mid-urethra, proximal at bladder neck, placed laterally.
4. Pass through Cooper's ligament; tie with vaginal elevation, without tension; leave about **two fingerbreadths** between bladder neck and pubis.
5. **Cystoscopy:** ureteric efflux, no suture in bladder.

- Laparoscopic and robotic versions match open short-term results.

Outcomes

- Cure 85–90% at 1 year, about 70% at 5 years (Cochrane).
- **SISTEr (2007):** autologous fascial sling more effective but more voiding dysfunction and UTI.
- **Ward and Hilton:** TVT (tension-free vaginal tape) equivalent to colposuspension; meta-analyses favour midurethral slings.
- **CARE (2006):** prophylactic Burch at sacrocolpopexy reduced postoperative SUI.

Complications

- Voiding dysfunction 2–27%, de novo storage symptoms 8–27%, **de novo prolapse** 2.5–27%; bladder or ureteric injury; bleeding.
- **Also know:** MMK (Marshall–Marchetti–Krantz; osteitis pubis 0.7–2.5%), paravaginal repair, VOS (vagino-obturator shelf); failure predictors: BMI (body mass index) >25, age >60, prior surgery.

Diagram to draw: sagittal pelvis with sutures from paravaginal fascia to Cooper's ligament.

Key points to remember:

- Burch 1961: vaginal wall to Cooper's ligament; for hypermobility.
- Cure 85–90% at 1 year, ~70% at 5 years.
- Niche: concomitant abdominal surgery, mesh avoidance.

Q3c. Drug Fosfomycin

Type: SAQ (10 marks)

Introduction

- Bactericidal **phosphonic acid derivative** antibiotic, chemically unrelated to other classes. Oral form: **fosfomycin trometamol**; intravenous form: fosfomycin disodium.

Mechanism of action

- Enters bacteria through glycerol-3-phosphate and glucose-6-phosphate transporters.
- Irreversibly inhibits **MurA** (enolpyruvyl transferase), blocking the **first step of peptidoglycan (cell wall) synthesis**. β-lactams and vancomycin act at later steps, so **cross-resistance is limited**.

Pharmacokinetics

- Oral bioavailability is modest, but the drug is excreted **unchanged in urine**. One 3 g dose keeps high urinary concentrations for about 1–2 days, which permits single-dose therapy. Low tissue levels.
- Concentrations fall in renal failure; avoid at very low creatinine clearance.

Spectrum

- **Escherichia coli**, including **ESBL (extended-spectrum β-lactamase)** producers; most Enterobacteriaceae (Klebsiella variable); *Enterococcus* including **VRE (vancomycin-resistant enterococcus)**; staphylococci.
- **Not reliable:** *Pseudomonas aeruginosa*, *Acinetobacter*.

Uses in urology

Indication	Dose
Uncomplicated cystitis in women (first-line, EAU (European Association of Urology) and IDSA (Infectious Diseases Society of America))	3 g oral sachet, single dose
Cystitis in pregnancy, asymptomatic bacteriuria (safe, category B)	3 g single dose
MDR (multidrug-resistant) lower UTI (urinary tract infection)	Oral option avoiding IV (intravenous) therapy; repeated doses used off-label
Prophylaxis for transrectal prostate biopsy	As effective as fluoroquinolone
Recurrent UTI prophylaxis	3 g every 10 days (EAU)

- **Not for pyelonephritis** or urosepsis orally (poor tissue levels). Emerging data suggest IV fosfomycin may help in bacteraemic MDR E. coli UTI.
- Not needed as prophylaxis for flexible cystoscopy.

Adverse effects

- Diarrhoea, GI (gastrointestinal) upset, headache, vaginitis; rare hypersensitivity. IV form: **sodium load**, hypokalaemia.

Resistance

- Transporter mutations, MurA changes, plasmid **fosA** enzymes. Resistance develops quickly in vitro, so systemic infections use combinations.

Evidence

- Huttner (2018) RCT (randomised controlled trial): nitrofurantoin 5 days gave higher clinical resolution than single-dose fosfomycin (70% vs 58%).

Diagram to draw: peptidoglycan synthesis showing fosfomycin (MurA), β-lactams and vancomycin sites.

> **Key points to remember:**
>
> - Inhibits MurA: first step of cell wall synthesis; no cross-resistance.
> - 3 g single dose for cystitis; active against ESBL E. coli and VRE.
> - Not for pyelonephritis; poor Pseudomonas cover.

Q3d. Use of contrast agents in urology and their complications

Type: SAQ (10 marks) (also answered in 2016-Summer Paper 1, Q1 and 2018-Summer Paper 1, Q3a)

Types and uses

Agent	Principle	Urological use
Iodinated: ionic high-osmolar (diatrizoate); non-ionic low-osmolar (iohexol); iso-osmolar dimer (iodixanol, 290 mOsm/kg)	Iodine absorbs X-rays; glomerular filtration	CT (computed tomography) urography, angiography; retrograde pyelogram, urethrogram, cystogram
Gadolinium (GBCA, gadolinium-based contrast agent)	Shortens T1 relaxation; chelated	MRI (magnetic resonance imaging) of renal mass, prostate, MR urography
Microbubbles	Gas in shell, intravascular	CEUS (contrast-enhanced ultrasound): cysts; voiding urosonography
Carbon dioxide	Negative contrast	Angiography in CKD (chronic kidney disease)

Complications of iodinated contrast

1. **Hypersensitivity:** immediate (within an hour: urticaria, bronchospasm, anaphylaxis) or non-immediate (hours to days, rash). Severe: 0.22% ionic vs 0.04% non-ionic. Risk: prior reaction, asthma, atopy. **Shellfish allergy is not a risk.**
2. **CA-AKI (contrast-associated acute kidney injury):** AKI within 48 hours (KDIGO (Kidney Disease: Improving Global Outcomes): creatinine rise ≥0.3 mg/dL or ≥1.5 × baseline). Medullary hypoxia plus tubular toxicity; real risk mainly at eGFR (estimated GFR) <30.
3. **Metformin:** lactic acidosis only if AKI; withhold 48 hours if eGFR <30 or AKI.
4. **Also:** physiological (warmth, vasovagal), thyrotoxicosis, extravasation. Retrograde studies are safe in allergy.

Complications of gadolinium

- **NSF (nephrogenic systemic fibrosis)** at GFR <30; group II agents carry almost no risk. Brain deposition with linear agents.

Prevention and treatment

- **Premedication** (ACR (American College of Radiology) 2023): prednisone 50 mg at 13, 7 and 1 hour, or methylprednisolone 32 mg at 12 and 2 hours, plus diphenhydramine 50 mg 1 hour before.
- CA-AKI: lowest dose, IV (intravenous) saline if eGFR <30; **PRESERVE (2018)**: bicarbonate and N-acetylcysteine no benefit.

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Stop injection → airway, oxygen
Anaphylaxis → IM (intramuscular) adrenaline 0.3–0.5 mg (max 1 mg); IV fluids
Bronchospasm → salbutamol; urticaria → H1 antihistamine
Observe ≥30 minutes
```

Diagram to draw: monomer versus dimer on an osmolality scale.

> **Key points to remember:**
> - Non-ionic agents: severe reactions 0.04% vs 0.22%.
> - Anaphylaxis: IM adrenaline 0.3–0.5 mg.
> - NSF risk at GFR <30; group II agents almost none.

Q3e. 46-XX Male

Type: SAQ (10 marks)

Assumed meaning: 46,XX testicular DSD (difference of sex development), "XX male" or de la Chapelle syndrome.

Definition

- Testes and male genitalia in a person with a 46,XX karyotype. Old term "XX sex reversal"; renamed **46,XX testicular DSD** (2006 consensus). First described by **de la Chapelle (1964)**. About **1 in 20,000** males.

Genetics

Type	Mechanism	Phenotype
SRY-positive (about 80–90%)	**SRY** (sex-determining region Y) translocated from Yp to the X chromosome during paternal meiosis	Usually normal male genitalia
SRY-negative	Testis genes switched on without SRY: **SOX9** duplication or upregulation, SOX3, loss of RSPO1 or WNT4	More hypospadias, undescended testes, ovotestis

- SRY activates SOX9, so Sertoli cells form and secrete **AMH (anti-Müllerian hormone)**: no uterus. Leydig testosterone virilises.
- The **AZF (azoospermia factor) region of Yq is absent**, so spermatogenesis fails.

Clinical features

- Normal male identity; often found at **infertility work-up** for azoospermia, or prenatally (cell-free DNA 46,XX but male ultrasound).
- Small firm testes, **azoospermia**, hypergonadotropic hypogonadism, gynaecomastia in some; progressive testicular regression.
- Height normal or **shorter than average** (no Y growth genes), unlike Klinefelter.

Feature	46,XX male	Klinefelter (47,XXY)
Frequency	1 in 20,000	1 in 500–1000
Height	Normal or short	Often tall
Genital anomalies	More common	Rare
Sperm retrieval	Never (no AZF)	Possible by micro-TESE (microdissection testicular sperm extraction)

Evaluation

- Semen analysis (azoospermia); FSH (follicle-stimulating hormone) and LH (luteinising hormone) high, testosterone low or low-normal.
- **Karyotype** 46,XX; **FISH (fluorescence in situ hybridisation) for SRY**; DSD gene panel if SRY-negative; scrotal ultrasound.

Management

- **Testosterone replacement** when hypogonadal (libido, bone density, muscle).
- **Infertility:** testicular sperm extraction is futile; counsel for **donor sperm or adoption**.
- Gynaecomastia: reduction mammoplasty if persistent; hypospadias or undescended testis repaired.
- Genetic counselling and psychological support; bone density monitoring.

Diagram to draw: X and Y chromosomes showing Xp–Yp exchange carrying SRY onto X.

Key points to remember:

- 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.
- Azoospermia always (AZF absent): no sperm retrieval.
- Shorter than Klinefelter; FISH for SRY confirms.

Q3f. Millin's prostatectomy

Type: SAQ (10 marks) (also answered in 2015-Summer Paper 2, Q3c)

Definition

- **Retropubic transcapsular simple prostatectomy** (Millin, 1945): adenoma enucleated through a transverse anterior capsulotomy **without opening the bladder**. Plane: the surgical capsule (compressed peripheral zone around the transition-zone adenoma).

Indications

- BPH (benign prostatic hyperplasia), gland **>80–100 g**, with refractory LUTS (lower urinary tract symptoms), retention, recurrent UTI (urinary tract infection), haematuria or upper-tract damage.
- Prefer suprapubic (Freyer) for a large median lobe, diverticulum or bladder stones.
- EAU (European Association of Urology) and AUA (American Urological Association): simple prostatectomy or HoLEP (holmium laser enucleation) for large glands.

Work-up

- PSA (prostate-specific antigen), MRI (magnetic resonance imaging) ± biopsy; TRUS (transrectal ultrasound) volume; cystoscopy; urodynamics if bladder function is doubtful.

Steps

1. Lower midline or Pfannenstiel; **space of Retzius**.
2. Optional prophylactic ligation of DVC (dorsal venous complex) and pedicles (Walsh–Oesterling).
3. **Transverse capsulotomy ~2 cm distal to the bladder neck** (a longitudinal cut may extend into the sphincter).
4. Find the adenoma–capsule plane with scissors; **finger enucleation**.
5. **Divide the apical urethra under vision** proximal to the sphincter.
6. **Figure-of-eight sutures at 5 and 7 o'clock** (prostatic arteries); advance bladder neck mucosa into the fossa.
7. 22 Fr three-way catheter, **watertight capsule closure**, drain.

Millin versus Freyer

Aspect	Millin (retropubic)	Freyer (suprapubic)
Bladder	Not opened	Opened
Apex	Divided under vision	Pinched by finger
Risk	Venous bleeding	Cystotomy leak

Complications

- Bleeding (transfusion <5%): catheter traction, Malament purse-string, O'Conor plication.
- Urine leak, UTI, bladder neck contracture 2–6%, stricture, DVT (deep vein thrombosis).
- **Retrograde ejaculation 80–90%**; ED (erectile dysfunction) 3–5%; incontinence very rare.

Advantages over TURP (transurethral resection of prostate)

- Complete removal, low retreatment, no TUR syndrome; but incision, longer stay.
- **Recent:** laparoscopic, robotic multiport and single-port transvesical simple prostatectomy: less bleeding, shorter stay.

Diagram to draw: sagittal view of transverse capsulotomy and enucleation plane.

Key points to remember:

- Millin 1945: capsulotomy ~2 cm below bladder neck; bladder not opened.
- Apex divided under vision; 5 and 7 o'clock sutures.
- Retrograde ejaculation 80–90%; transfusion <5%.

Rapid Revision Sheet – 2022-Summer Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe anatomy of kidney in relation to PCNL and describe indication, contraindication, procedure and complications of mini-PCNL. (LAQ)

- Fornix puncture: no arterial injury; upper-pole infundibulum 67% vascular injury.
- Pleura to 12th rib; retrorenal colon at the lower pole, commoner prone.
- Mini 15–22 Fr (Jackman 1998): fewer transfusions, longer operation.
- High intrarenal pressure in mini-PCNL causes fever and sepsis.

Q2. Urological evaluation and management of pediatric kidney transplant patient. (LAQ)

- Bladder target: age-expected capacity at <40 cm H_2O; augment if <75% despite CIC.
- PUV bladder heads to myogenic failure; neuropathic bladder needs earlier reconstruction.
- Never test a bladder with a graft; cycle the defunctionalised bladder.
- Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.

Q3a. Instestinal segments in urology (SAQ)

- Ileum and colon: hyperchloraemic acidosis (NH_4Cl absorption).
- Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
- Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.

Q3b. Burch colposuspension (SAQ)

- Burch 1961: vaginal wall to Cooper's ligament; for hypermobility.
- Cure 85–90% at 1 year, ~70% at 5 years.
- Niche: concomitant abdominal surgery, mesh avoidance.

Q3c. Drug Fosfomycin (SAQ)

- Inhibits MurA: first step of cell wall synthesis; no cross-resistance.
- 3 g single dose for cystitis; active against ESBL E. coli and VRE.
- Not for pyelonephritis; poor Pseudomonas cover.

Q3d. Use of contrast agents in urology and their complications (SAQ)

- Non-ionic agents: severe reactions 0.04% vs 0.22%.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg.
- NSF risk at GFR <30; group II agents almost none.

Q3e. 46-XX Male (SAQ)

- 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.
- Azoospermia always (AZF absent): no sperm retrieval.
- Shorter than Klinefelter; FISH for SRY confirms.

Q3f. Millin's prostatectomy (SAQ)

- Millin 1945: capsulotomy ~2 cm below bladder neck; bladder not opened.
- Apex divided under vision; 5 and 7 o'clock sutures.
- Retrograde ejaculation 80–90%; transfusion <5%.